

7(a)(i)	non-inverting (amplifier)	B1
7(a)(ii)	$\text{gain} = \frac{R_f}{R} + 1$ $\text{gain} = \frac{3.6}{0.72} + 1 = 6.0$	B1
7(a)(iii)	straight line from (0,0) to ($T/2$, 3)	B1
	line from origin to 3.0 V then horizontal line at 3.0 V to T	B1
7(a)(iv)	ldr / light dependent resistor replaces one of the two resistors	B1
7(b)(i)	relay coil	B1
7(b)(ii)	relay coil between op-amp and earth	B1
	diode with correct polarity (pointing away from output) connected between output and device and no other connections or diode with correct polarity (pointing towards earth) between device and earth and no other connections	B1
	switch connected to high voltage circuit	B1

Question	Answer	Marks
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